

Hunmin Yang

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Personal Website

Assistant Professor @ Korea University Sejong Campus

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RESEARCH INTEREST

- I am broadly interested in machine learning and computer vision, with a recent focus on security and safety aspects of Generative AI, Physical AI, and Agentic AI. The goal of my research is to develop trustworthy multimodal intelligence to enable safe and reliable model deployment in real-world applications. To achieve this, I work at the intersection of AI and security. I hope to continue making a positive impact on society by enhancing the trustworthiness of AI.

Keywords: *AI Security, AI Safety, Trustworthy AI, Multiodal AI*

EXPERIENCE

- **Korea University Sejong Campus** Sejong, Korea
Assistant Professor *Sep 2026 - Present*
 - Leading Trustworthy Multimodal AI (TMAI) Lab
- **Agency for Defense Development (ADD)** Daejeon, Korea
Senior AI Researcher *Jan 2020 - Aug 2026*
 - **GAID**: Multi-modal LLMs & diffusion models for generative AI
 - **D-CAM**: Adversarial attack & defense techniques for robust AI
- **Agency for Defense Development (ADD)** Daejeon, Korea
AI Researcher *May 2017 - Dec 2019*
 - **D-GEN**: Synthetic data generation framework for training AI
 - **D-NET**: Large-scale AI inference platform with Hadoop-Spark
- **Agency for Defense Development (ADD)** Daejeon, Korea
Specialized Research Staff (Military Service) *Feb 2014 - May 2017*

EDUCATION

- **Korea Advance Institute of Science and Technology (KAIST)** Daejeon, Korea
PhD in Mechanical Engineering *Sep 2021 - Feb 2026*
 - Dissertation: “Generative Model-based Transferable Attacks on Deep Vision Models”
 - Research Area: Adversarial Machine Learning
 - Advisor: Kuk-Jin Yoon
- **Korea Advance Institute of Science and Technology (KAIST)** Daejeon, Korea
MS in Mechanical Engineering *Feb 2012 - Feb 2014*
 - Thesis: “Virtual Placement of 3D Sound Source at Upper Hemisphere using Ear-level Loudspeakers”
 - Research Area: 3D Sound Perception
 - Advisor: Youngjin Park
- **Royal Melbourne Institute of Technology (RMIT)** Melbourne, Australia
Exchange Student (High Distinction) *Feb 2011 - Aug 2011*
- **Korea Advance Institute of Science and Technology (KAIST)** Daejeon, Korea
BS in Mechanical Engineering (Magna Cum Laude) *Feb 2007 - Feb 2012*

PUBLICATIONS

1. Jongoh Jeong, **Hunmin Yang**, Jaeseok Jeong, Kuk-Jin Yoon. “Improving Black-Box Generative Attacks via Generator Semantic Consistency.” In *International Conference on Learning Representations (ICLR)*, 2026.
2. Junhyeong Cho, Kim Youwang, **Hunmin Yang**, Tae-Hyun Oh. “Robust 3D Shape Reconstruction in Zero-Shot from a Single Image in the Wild.” In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
3. Jongoh Jeong, **Hunmin Yang**, Kuk-Jin Yoon. “Boosting Adversarial Transferability with a Generative Model Perspective.” In *CVPR Workshop on Generative Models for Computer Vision (GMCV)*, 2025.
4. Se-Yoon Oh, **Hunmin Yang**. “Physical Adversarial Attacks and Defenses on Vehicle Detection Models.” In *International Conference on Control, Automation and Systems (ICCAS)*, 2025.
5. **Hunmin Yang**, Jongoh Jeong, Kuk-Jin Yoon. “Prompt-Driven Contrastive Learning for Transferable Adversarial Attacks.” In *European Conference on Computer Vision (ECCV)*, 2024. (**Oral, top 2.3%**)
6. Junhyeong Cho, Kim Youwang, **Hunmin Yang**, Tae-Hyun Oh. “Object-Centric Domain Randomization for 3D Shape Reconstruction in the Wild.” In *CVPR Workshop on Foundation Models (WFM)*, 2024.
7. **Hunmin Yang***, Jongoh Jeong*, Kuk-Jin Yoon. “FACL-Attack: Frequency-Aware Contrastive Learning for Transferable Adversarial Attacks.” In *Association for the Advancement of Artificial Intelligence (AAAI)*, 2024.
8. Junhyeong Cho, Gilhyun Nam, Sungyeon Kim, **Hunmin Yang**, Suha Kwak. “PromptStyler: Prompt-driven Style Generation for Source-free Domain Generalization.” In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
9. Naufal Suryanto, Yongsu Kim, Harashta Tatimma Larasati, Hyeon Kang, Thi-Thu-Huong Le, Yoonyoung Hong, **Hunmin Yang**, Se-Yoon Oh, Howon Kim. “ACTIVE: Towards Highly Transferable 3D Physical Camouflage for Universal and Robust Vehicle Evasion.” In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
10. **Hunmin Yang**, Se-Yoon Oh, Junhyeong Cho. “Synthetic Image Generation for Deep Neural Networks.” In *NVIDIA GPU Technology Conference (GTC)*, 2023. (**Showcase**)
11. Se-Yoon Oh, **Hunmin Yang**. “Simulation of Physical Adversarial Attacks on Vehicle Detection Models.” In *International Conference on Control, Automation and Systems (ICCAS)*, 2023.
12. Naufal Suryanto, Yongsu Kim, Hyeon Kang, Harashta Tatimma Larasati, Youngyeo Yun, Thi-Thu-Huong Le, **Hunmin Yang**, Se-Yoon Oh, Howon Kim. “DTA: Physical Camouflage Attacks using Differentiable Transformation Network.” In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
13. Jeonghun Kim, Kyungmin Lee, Hyeongkeun Lee, **Hunmin Yang**, Se-Yoon Oh. “Camouflaged Adversarial Attack on Object Detector.” In *21th International Conference on Control, Automation and Systems (ICCAS)*, 2021.
14. **Hunmin Yang**, Se-Yoon Oh, Taewon Kim, Ki-Jung Ryu. “D-GEN: A Deep Learning Data Generation Framework For Artificial Intelligence.” In *NVIDIA GPU Technology Conference (GTC)*, 2020.
15. Kyungmin Lee, **Hunmin Yang**, Se-Yoon Oh. “Adversarial Training on Joint Energy Based Model for Robust Classification and Out-of-Distribution Detection.” In *International Conference on Control, Automation and Systems (ICCAS)*, 2020.
16. Eunchong Kim, Kanghyun Park, **Hunmin Yang**, Se-Yoon Oh. “Training Deep Neural Networks with Synthetic Data for Off-Road Vehicle Detection.” In *International Conference on Control, Automation and Systems (ICCAS)*, 2020.
17. Hyeongkeun Lee, Kyungmin Lee, **Hunmin Yang**, Se-Yoon Oh. “Applying FastPhotoStyle to Synthetic Data for Military Vehicle Detection.” In *International Conference on Control, Automation and Systems (ICCAS)*, 2020.
18. Kanghyun Park, Hyeongkeun Lee, **Hunmin Yang**, Se-Yoon Oh. “Improving Instance Segmentation using Synthetic Data with Artificial Distractors.” In *International Conference on Control, Automation and Systems (ICCAS)*, 2020.
19. **Hunmin Yang**, Se-Yoon Oh, Ki-Jung Ryu. “Accelerating Distributed Deep Learning Inference on multi-GPU with Hadoop-Spark.” In *NVIDIA GPU Technology Conference (GTC)*, 2019. (**Oral**)
20. **Hunmin Yang**, Se-Yoon Oh, Ki-Jung Ryu. “Scalable Distributed Deep Learning Inference on Multi-GPU with Hadoop-Spark.” In *NVIDIA GPU Technology Conference (GTC)*, 2019.
21. Se-Yoon Oh, **Hunmin Yang**, Ki-Jung Ryu. “Optimal Distributed Inference on Multi-GPU Processing System.” In *NVIDIA GPU Technology Conference (GTC)*, 2019.

22. Se-Yoon Oh, **Hunmin Yang**, Ki-Jung Ryu. “Optimal Experimental Design Approach for Machine Learning Process.” In *International Conference on Control, Automation and Systems (ICCAS)*, 2017.

PATENTS

Machine Learning & Synthetic Data Generation

- **Hunmin Yang**, Se-Yoon Oh. Method and apparatus for generating training data for deep learning models. kr 10-2613781, 2023.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Apparatus and method for deep learning based on a hybrid of virtual and real data. kr 10-2198088, 2020.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Apparatus and method for training machine learning models using virtual data. kr 10-2086351, 2020.
- **Hunmin Yang**, Se-Yoon Oh, Seongbaek Jo. Apparatus and method for enhancing learning performance in machine learning. kr 10-2053202, 2019.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Method and apparatus for improving self-supervised learning performance using synthetic data. kr 10-2032519, 2019.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Method and apparatus for adding artificial objects to improve object detection performance. kr 10-1972095, 2019.
- **Hunmin Yang**, Se-Yoon Oh, Seongbaek Jo. Apparatus and method for generating training images based on game engines for machine learning. kr 10-1947650, 2019.

AI Security & Adversarial Robustness

- Se-Yoon Oh, **Hunmin Yang**, Hyeongkeun Lee, Kyungmin Lee, Jeonghun Kim. Method and apparatus for optimizing adversarial patches, computer-readable storage medium, and computer program. kr 10-2445215, 2022.
- Hyeongkeun Lee, Jeonghun Kim, Kyungmin Lee, **Hunmin Yang**, Se-Yoon Oh. Method and apparatus for optimizing adversarial patches, computer-readable storage medium, and computer program. kr 10-2414146, 2022.
- Jeonghun Kim, Se-Yoon Oh, Hyeongkeun Lee, Kyungmin Lee, **Hunmin Yang**. Apparatus and method for optimizing natural-pattern-based adversarial patches for stealthiness against human vision. kr 10-2380154, 2022.
- Hyeongkeun Lee, **Hunmin Yang**, Jeonghun Kim, Kyungmin Lee, Se-Yoon Oh. Method, apparatus, computer-readable storage medium, and computer program for determining adversarial patch positions. kr 10-2360070, 2022.

Big Data & Database

- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Method and apparatus for building NoSQL databases for signal processing. kr 10-2002360, 2019.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Method and apparatus for building inverse index databases for high-speed video object retrieval. kr 10-2014267, 2019.

HONORS AND AWARDS

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| <ul style="list-style-type: none"> • Defense Science Award
 <i>From the Chief Research Director of ADD</i> <ul style="list-style-type: none"> ◦ Adversarial Attack Countermeasures for Robust AI-based Image Analysis Systems • Selected as Oral Presentation (top 2.3%)
 <i>In European Conference on Computer Vision (ECCV)</i> <ul style="list-style-type: none"> ◦ Prompt-Driven Contrastive Learning for Transferable Adversarial Attacks • National Grant for Defense Research and Development
 <i>From the Chief Director of DAPA</i> <ul style="list-style-type: none"> ◦ Synthetic Data Generation for Defense AI • Defense Science Award
 <i>From the Chief Research Director of ADD</i> <ul style="list-style-type: none"> ◦ Improving Distributed Multi-GPU Computing for Large-scale Video Analytics | <div>Aug 2025</div> <div>Oct 2024</div> <div>Dec 2021</div> <div>Aug 2019</div> |
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- **High Achievement Award** Aug 2018
From the Chief Research Director of ADD
 - Big Data Platform Development and Synthetic Data Generation
- **Excellent Paper Award** Mar 2013
From the Korea Society for Noise and Vibration Engineering (KSNVE)
 - Sweet Spot Analysis of Linear Array System by Geometrical Approach
- **Scholarship for Academic Excellence** Jun 2007
From the Korea Human Resource Development Scholarship Association
 - Outstanding Academic Performance
- **Scholarship for Academic Excellence** 2007-2014
From the Korean Government
 - Tuition free for all semesters in KAIST (BS & MS)

PROFESSIONAL SERVICES

- **Conference Reviewer**
 - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
 - IEEE/CVF International Conference on Computer Vision (ICCV)
 - European Conference on Computer Vision (ECCV)
 - Neural Information Processing Systems (NeurIPS)
 - AAAI Conference on Artificial Intelligence (AAAI)
 - ACM Multimedia Conference (MM)
 - Association for Computational Linguistics (ACL)
- **Journal Reviewer**
 - International Journal of Computer Vision (IJCV)
 - IEEE Transactions on Dependable and Secure Computing (TDSC)
- **Technology Transfer**
 - Synthetic data generation for training AI models → SI Analytics, Jcorp System, JEIOS
 - Physical adversarial camouflage generation for attacking AI models → SmartM2M
 - Big data platform for large-scale AI model inference → XIIIlab

REFERENCES

- **Kuk-Jin Yoon** (PhD Advisor) kjyoon@kaist.ac.kr
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Professor, Department of Mechanical Engineering, KAIST
- *Deputy Director, KAIST AI Institute*
Interim Dean, College of AI, KAIST
- **Youngjin Park** (MS Advisor) yjpark@kaist.ac.kr
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Professor, Department of Mechanical Engineering, KAIST